

3 Partial Derivatives

3.4 Tangent Planes and Linear Approximations

Example 3.4.1 Find the tangent plane to the elliptic paraboloid

$$z = 2x^2 + y^2$$

at the point $(1, 1, 3)$.

The tangent plane can be used to approximate values of a function near a point whose value is known:

Example 3.4.2 Use the tangent plane to approximate $f(1.1, -0.1)$ where

$$f(x, y) = xe^{xy}.$$

Example 3.4.3 Use a tangent plane to approximate the value of $\sqrt{3.05^2 + 3.97^2}$.

What happens when the partial derivatives are not continuous at the point (a, b) ?

Example 3.4.4 Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = 0. \end{cases}$$

Find the first partial derivatives of f at the point $(0, 0)$. What would the linearization of f be at the origin? Is this a good approximation?

Linear approximations really only make sense for differentiable functions.

Definition 3.4.5 Let $\Delta y = f(a + \Delta x) - f(a)$. The function $f(x)$ is said to be differentiable at a if there exists a function ϵ such that

$$\Delta y = f'(a)\Delta x + \epsilon\Delta x$$

where $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$.

For functions of two variables, we have

Definition 3.4.6 Let $\Delta z = f(a + \Delta x, b + \Delta y) - f(a, b)$. The function $f(x, y)$ is said to be differentiable at (a, b) if there exists a functions ϵ_1 and ϵ_2 such that

$$\Delta z = f_x(a, b)\Delta x + f_y(a, b)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y$$

where $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$.

Theorem 3.4.7 If f_x and f_y both exist near (a, b) and are continuous at (a, b) , then f is differentiable at the point (a, b) .

Example 3.4.8 The base radius and height of a right circular cone are measured as 10 cm and 25 cm, respectively, with a possible error in measurement of as much as 0.01 cm in each. Use a tangent plane to estimate the maximum error in the calculated volume of the cone.